

A techno-economic assessment of carbon dioxide removal pathways via biochemical conversion of lignocellulose to biofuels and bioplastics

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ABSTRACT

Biomass carbon removal and storage (BiCRS) is a promising pathway to mitigate climate change via large scale removal of atmospheric carbon dioxide (CO₂). We modeled several fermentation technologies, producing a variety of bioproducts from lignocellulosic feedstocks, to understand their levelized cost of CO₂ removal under multiple scenarios. Lifecycle greenhouse gas (GHG) emissions are accounted to provide cradle-to-grave estimates of carbon intensity (CI). We did not account for the avoided fossil CO₂ emissions from the use of biofuels in our CO₂ removal cost calculations, because avoided emissions do not contribute to CO₂ removal. The main products from the fermentation technologies we modeled include renewable diesel, ethanol, sustainable aviation fuel (SAF), and polyethylene (PE), with co-products including CO₂, adipic acid, steam, and electricity. PE, depending on its end-of-life management, can serve as a form of biogenic carbon storage. PE has the potential to remove 1.2–1.5 tCO₂ per dry t-biomass, whereas biofuels have the potential to remove 0.3–0.9 tCO₂ per dry t-biomass, indicating that PE production is a more efficient method of carbon removal. We quantify costs of CO₂ removal to be \$60 – \$675 per metric tCO₂ removed across the various fermentation pathways. Under the scenarios analyzed, bioplastic production from lignocellulosic biomass is a more cost-effective route to CO₂ removal than biofuel production, with costs of CO₂ removal via bioplastics being 50–90 % lower than that of biofuels. Future research should explore the potential benefits and drawbacks of expanding bioplastic production for large-scale CO₂ removal.

1. Introduction

The conversion of renewable biomass resources into valued bio-products via biorefining can enhance energy system sustainability, reduce greenhouse gas emissions by displacing fossil fuels, and boost local economies by increasing domestic jobs, to name a few [1–3]. This multifaceted potential has positioned biorefining at the forefront of global strategies to transition toward low-carbon, circular economies. In particular, biorefining contributes to decarbonization efforts by integrating waste valorization, energy production, and material synthesis into a unified framework that leverages regionally available biomass. In

doing so, biorefineries can reduce reliance on volatile international fuel markets while promoting energy independence and rural development [4,5]. Furthermore, biorefineries are increasingly viewed not only as alternatives to petroleum refineries but also as key enablers of sustainable industrial systems that align with emerging climate and resource conservation goals.

Biochemical methods of biorefining rely on the use of microorganisms to convert biomass into bioproducts via various bioprocesses including fermentation. These microbial platforms are engineered or selected for their ability to metabolize diverse carbon sources and synthesize specific target molecules, such as organic acids, alcohols, solvents, and polymers. Biochemical processing offers advantages over

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Abbreviations		Units and nomenclature	
ADA	Adipic acid	Al ₂ O ₃	Aluminum Oxide
BECCS	Bioenergy with carbon capture and storage	CH ₄	Methane
BiCRS	Biomass carbon removal and storage	CO ₂	Carbon dioxide
CCS	Carbon capture and storage	CO _{2e}	Carbon dioxide equivalent
CDR	Carbon dioxide removal	d	Days
CI	Carbon intensity	dry t	Dry metric tonne
DAC	Direct air capture	EtOH	Ethanol
DMR	Deacetylation and mechanical refining	g	Gram
GHG	Greenhouse gas	gal	Gallons
IPCC	Intergovernmental Panel on Climate Change	H ₂	Hydrogen
LC	Levelized cost	kg	Kilograms
LCA	life cycle assessment	kWh	Kilowatts-hour
MEA	Monoethanolamine	MW	Mega watts
NREL:	National Renewable Energy Laboratory	MJ	Mega Joules
PE	Polyethylene	NaOH	Sodium hydroxide
RD	Renewable diesel	N ₂ O	Nitrous oxide
SAF	Sustainable aviation fuel	t	Metric tonne
TCI	Total capital investment	tCO ₂	Metric tonne of CO ₂
TDC	Total direct costs	ZrO	Zirconium oxide
TEA	Techno-economic analysis		

thermochemical routes in terms of lower operating temperatures, higher selectivity, and compatibility with aqueous systems [6]. Additionally, advances in synthetic biology and metabolic engineering have enabled the development of designer microbes with enhanced tolerance, productivity, and substrate utilization, thereby improving process economics and reducing the carbon footprint of bio-based products [7]. Biochemical conversion is particularly suitable for biomass materials rich in cellulose and hemicellulose, such as lignocellulose, since these polymers can be enzymatically broken down into monomeric sugars and subsequently upgraded into bioproducts via fermentation [3]. Lignocellulosic feedstocks, which include agricultural residues, forestry by-products, and energy crops, are among the most abundant and underutilized renewable resources available globally. The efficient deconstruction of lignocellulose, however, remains a technical challenge due to the recalcitrance of lignin and the heterogeneous nature of plant biomass. Recent innovations in pretreatment technologies, enzyme engineering, and consolidated bioprocessing have made significant strides in overcoming these hurdles, thus opening the door to more widespread adoption of lignocellulosic biorefineries [8]. By enabling the valorization of non-food, lignocellulosic biomass, biochemical conversion also addresses concerns about food security and land use competition often associated with first-generation biofuels.

To date, the primary motive for advancing the state-of-the-art of biochemical conversion technologies has been biofuel production, but recently a new motive has emerged: biomass carbon removal and storage (BiCRS). Unlike conventional biofuel strategies, which aim to achieve carbon neutrality by offsetting fossil carbon emissions, BiCRS targets net-negative emissions by permanently sequestering atmospheric carbon absorbed by biomass during growth. This is achieved by integrating biomass conversion with carbon capture and long-term storage methods, such as biochar stabilization, mineralization, or geological injection of biogenic CO₂. Thus, BiCRS mitigates climate change by leveraging bioprocessing technologies to durably remove CO₂ from the atmosphere [9]. BiCRS has therefore emerged as a promising tool within the portfolio of carbon dioxide removal (CDR) strategies recommended by the Intergovernmental Panel on Climate Change (IPCC) and other climate policy bodies to meet global warming targets [10]. Furthermore, BiCRS can offer co-benefits beyond climate mitigation, including soil restoration, nutrient retention, and waste reduction, depending on the specific pathways employed. As such, integrating

BiCRS into biorefining systems presents an opportunity to simultaneously produce valuable bioproducts and contribute to global carbon management goals. Future research should focus on optimizing these dual objectives by advancing microbial chassis, improving carbon retention across the process chain, and developing robust lifecycle assessment frameworks tailored to carbon removal performance [11].

2. Biomass carbon removal and storage (BiCRS)

According to the IPCC, 5–10 gigatons of CO₂ must be removed from the atmosphere per year by mid-century if we are to avoid the worst effects of climate change [12]. This urgent requirement for CDR is in addition to, not instead of, global efforts to transition to low-carbon energy systems, implement deep emissions cuts across all sectors, and improve the resilience of ecosystems and societies to climate-related risks. Achieving such a massive scale of CDR will require an integrated portfolio of solutions, including both engineered and nature-based pathways, deployed across multiple geographies and value chains [13–15]. Prominent methods of technology-enabled CDR include direct air capture (DAC), BiCRS, and mineralization [9]. BiCRS involves the conversion of sustainably sourced biomass carbon, including lignocellulose, into valued bioproducts that durably remove CO₂ from the atmosphere over their life cycle. This approach leverages the natural carbon fixation of photosynthesis while engineering downstream processes to ensure long-term carbon storage, making BiCRS one of the few negative-emission strategies that can generate economic value while addressing climate change [16,17]. BiCRS technologies can be broadly classified as either thermochemical or biochemical; examples include (respectively) gasification to H₂ with sequestered CO₂ as a co-product, or fermentation to ethanol with sequestered CO₂ as a co-product (Fig. 1).

The distinction between these categories is largely defined by process conditions—high-temperature conversion versus biological catalysis—and each has its own set of opportunities and technical challenges. Thermochemical pathways often benefit from higher conversion efficiencies and have reached greater commercial maturity, while biochemical pathways offer selectivity, milder processing conditions, and potential for integration with renewable electricity and green hydrogen systems [18].

Generally, three thermochemical pathways have received considerable attention in the field of BiCRS: 1) biomass combustion to produce

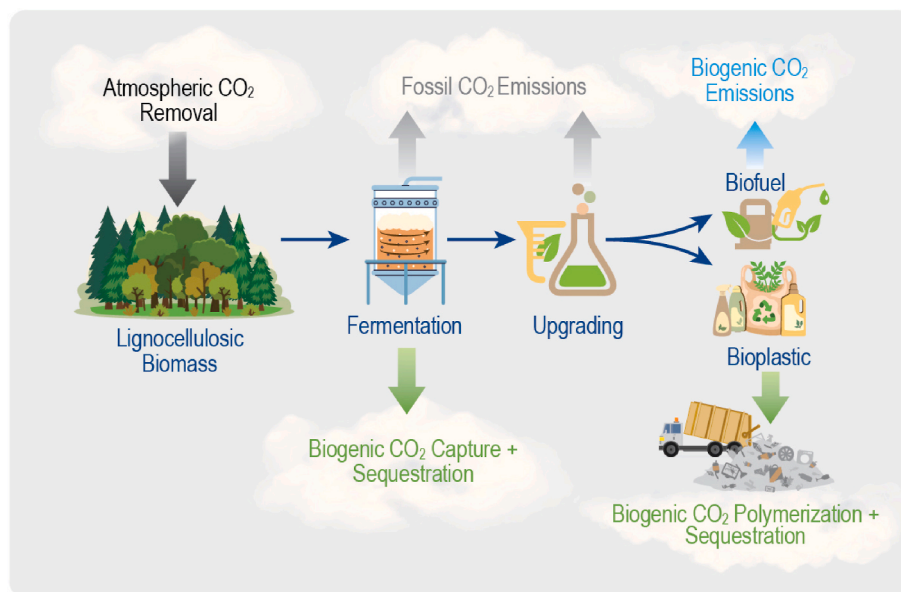


Fig. 1. A carbon-negative biorefinery scheme involving fermentation of lignocellulosic biomass into biofuel or bioplastic with CO₂ capture and sequestration.

heat/power and relatively concentrated CO₂ (10–15 %) for sequestration, 2) gasification to produce hydrogen (H₂) and highly concentrated CO₂ (>95 %) for sequestration, and 3) pyrolysis to produce biochar for sequestration [9]. These routes are attractive in part because they are compatible with lignocellulosic feedstocks and benefit from economies of scale. Moreover, advances in carbon capture technologies have improved the feasibility of coupling thermochemical conversion with CO₂ sequestration in both geological and mineralized forms [18]. The first pathway is commonly referred to as bioenergy with carbon capture and sequestration (BECCS) [19–22], and became a popular form of BiCRS in large part because of its prominent inclusion in reports published by the Intergovernmental Panel on Climate Change (IPCC) [23]. BECCS is often modeled as a cornerstone of 1.5°C-aligned climate scenarios because of its theoretical ability to produce energy while delivering net-negative emissions. BECCS has been slow to deploy in part due to its high cost of electricity relative to renewable energy solutions including solar photovoltaic and wind turbines [24]. Additionally, concerns about land use, water demand, and social acceptance have complicated its implementation, particularly in regions with competing food and energy needs [18,25]. Gasification is a promising method of BiCRS due to its ability to produce CO₂ of high purity as well as a high-value co-product in the form of hydrogen gas. Hydrogen derived from biomass has been proposed as a key vector in decarbonizing hard-to-electrify sectors such as steelmaking, aviation, and shipping, with the added benefit of enabling carbon-negative hydrogen production when paired with CCS [26]. Gasification with CCS has been slow to deploy in part due to its technical complexity and reliance on a strong and growing demand for hydrogen gas, which has yet to materialize [9]. Other barriers include the need for high capital investment, challenges in feedstock handling and reactor operation, and uncertainty around long-term policy support and market pricing for negative emissions credits [26,27].

Pyrolysis to biochar is another promising method of BiCRS due to its relative simplicity and modularity [9]. Biochar can be applied to soils to enhance nutrient retention, water holding capacity, and microbial activity, offering potential co-benefits in agricultural systems [24]. Relative to BECCS and gasification, pyrolysis is growing more rapidly but suffers from uncertainty around the permanence of biochar and lack of high-value products [24]. The long-term stability of biochar carbon in diverse soil environments remains uncertain and is influenced by factors such as feedstock type, pyrolysis conditions, and soil characteristics

[28–30]. Efforts are ongoing to expand the suite of biochar-based products, such as composites, building materials, and activated carbon, to improve the economics of pyrolysis-based BiCRS systems [31]. Biochemical pathways for carbon removal, including fermentation, have not been explored to the same extent as thermochemical pathways for reasons that are unclear. One plausible explanation is that research and policy attention have historically prioritized energy-centric thermochemical systems that align with fossil fuel substitution narratives, while biochemical systems—typically associated with biofuel production—have not been viewed through a carbon removal lens. Innovators in the fields of clean energy and greenhouse gas mitigation have generally developed fermentation technologies for the production of biofuels, which doesn't allow for significant carbon removal since most of the biogenic carbon is emitted back into the atmosphere upon combustion. Another potential reason could be that the aforementioned thermochemical pathways have all been proven at commercial scale to be effective at converting non-food, sustainable, and scalable biomass feedstocks, such as lignocellulose, into bioproducts, whereas conversion of these feedstocks via fermentation is at an earlier stage of development. However, recent innovations in fermentation-based systems offer new pathways for durable carbon retention through the synthesis of stable molecules or materials with long product lifetimes [31]. The sum of CO₂ captured and sequestered must exceed the emissions (CO_{2e}) over the life cycle of the bioproduct to result in atmospheric CO₂ removal and be considered carbon-negative [32]. There is a strong need to better understand the techno-economic feasibility of large-scale CO₂ removal via fermentation technologies. Future work should also assess policy frameworks, lifecycle carbon accounting, and supply chain design to support commercialization of biochemical BiCRS technologies alongside thermochemical approaches [33].

2.1. Carbon-negative fermentation: Opportunities and challenges

The conversion of biomass into biofuels, biochemicals, and biomaterials through fermentation is a key sector in the global bioeconomy. As countries aim to meet climate targets while ensuring energy security and economic resilience, fermentation-based bioprocessing is emerging as a flexible and scalable technology capable of producing both energy carriers and value-added chemicals with the potential for net-negative emissions when paired with carbon capture and storage (CCS). Ethanol and biomethane are two biofuels generated in large quantities

via fermentation technologies, both of which generate CO₂-rich exhaust streams, often exceeding 90 % purity, making them well-suited for cost-efficient CO₂ capture and sequestration [34–37]. These gases, when captured, can be compressed and stored geologically or valorized into stable carbon-based products, depending on the application and infrastructure available [38]. This makes fermentation-based biofuel systems unique among renewable energy technologies, as they enable co-production of both energy and a variety of durable carbon sinks. In the U.S., leveraging existing ethanol and biomethane industries could facilitate the capture of approximately 45 million and 110 million metric tonnes of biogenic CO₂ per year, respectively [39]. This scale of biogenic CO₂ generation is significant when compared to the total potential of other negative emissions technologies currently under development. If harnessed effectively, it has the potential to provide a near-term deployment pathway for large-scale carbon removal at relatively low marginal cost. Capturing 60% of CO₂ emissions from ethanol refineries could be achieved at costs below \$25 per metric tonne, leading to only a modest ~3.5% increase in ethanol fuel prices [39]. A practical example of large-scale CCS implementation is Archer Daniels Midland's successful injection of over one million metric tonnes of fermentation-derived CO₂ into geological storage in Illinois [40]. This commercial-scale demonstration proves the technical feasibility and regulatory viability of fermentation-based CCS in regions with favorable geology. Moreover, it offers a template for similar facilities worldwide, particularly in agricultural regions with high biomass output and proximity to sequestration formations. Biomethane from anaerobic digestion has gained popularity due to its ability to displace fossil natural gas in heating and transportation applications; however, the diffuse and decentralized nature of many anaerobic digestion systems presents challenges for integrating CCS cost-effectively [38]. Specifically, the variability in CO₂ capacity and quality across different feedstock sources—such as wastewater treatment plants, agricultural digesters, and landfills—complicates capture efforts. Advanced biogas separation technologies, including chemical absorption and pressure swing adsorption, can improve CO₂ extraction, though continued development is necessary for more cost-effective, modular systems [41,42]. Emerging technologies such as membrane-based gas separation, cryogenic upgrading, and biological methanation are being explored to reduce energy consumption and capital costs in small-scale applications [43,44]. A major concern in anaerobic digestion is methane leakage, given its significant global warming potential. To mitigate this, emerging technologies like dark fermentation, which produces hydrogen and CO₂ while eliminating methane emissions, are being explored as a viable alternative [45]. These technologies not only reduce fugitive methane emissions but also open new avenues for combining green hydrogen production with CO₂ capture, potentially aligning with hydrogen economy roadmaps [46]. Ethanol and biomethane are mostly derived from readily degradable biomass feedstocks, including starch and food waste, which present challenges to achieving large-scale CO₂ removal due to their high embodied emissions and/or low abundance [9]. These feedstocks suffer from regional availability constraints, which limit their scalability in global carbon removal scenarios. Lignocellulose is a promising biomass feedstock for fermentation with significant potential for large scale CO₂ removal due to its low embodied emissions and high abundance; examples of lignocellulose feedstocks include agricultural residues (e.g. corn stover, sugarcane bagasse), perennial grasses (e.g. switchgrass, miscanthus), and forestry residues [9]. These feedstocks have minimal land-use change impacts and often represent agricultural or industrial waste streams, making them ideal candidates for carbon-negative bioprocessing strategies [46]. The choice of biomass feedstock for carbon-negative fermentation is important considering the associated impacts on net carbon removal over the bioproduct life cycle. Advanced life-cycle assessment (LCA) methods are being developed to quantify these impacts and evaluate trade-offs related to land use, nutrient cycling, and biodiversity conservation [47]. The majority of studies on biomass fermentation with CCS do not quantify the costs of

atmospheric carbon removal by evaluating lifecycle GHG emissions but are rather focused on the cost of CO₂ capture alone. To better inform climate policy and investment strategies, it is essential to consider full system boundaries, including upstream emissions, feedstock logistics, energy inputs, and sequestration permanence [47]. Although lignocellulose holds great potential for large-scale carbon-negative fermentation, it requires bioprocess unit operations that are more complicated than those used for the processing of starch and other readily degradable feedstocks. Specifically, the fermentation of lignocellulosic biomass typically involves pretreatment of the biomass to remove lignin and increase the accessibility of cellulose and hemicellulose for enzymatic hydrolysis, followed by fermentation of the resulting sugars to bioproducts [48]. Common pretreatment approaches include dilute acid hydrolysis, steam explosion, and organosolv processes, all of which require careful optimization to balance sugar yield, enzyme accessibility, and inhibitor formation [49]. The development of lignocellulose fermentation technologies is still in its early stages, and several challenges (e.g. production costs, conversion yields, and integration strategies) need to be overcome to achieve commercial viability [50,51]. These challenges also include improving microbial tolerance to feedstock-derived inhibitors, optimizing co-fermentation of C5 and C6 sugars, and developing scalable enzyme production platforms. Although past large-scale attempts to ferment lignocellulosic biomass were unsuccessful, renewed interest in the process is driven by its ability to produce higher-value products such as polyethylene (PE) and adipic acid, while also capitalizing on CO₂ sequestration. Such co-product strategies increase the economic viability of fermentation-based BiCRS by offsetting the costs of CO₂ capture and storage with revenue from industrial chemicals and sustainable materials [52]. Long-term carbon-negative fermentation strategies could couple on-site CO₂ capture with on-field soil carbon sequestration through the use of perennial grasses, such as switchgrass, which have demonstrated climate benefits up to 15 times greater than grassland restoration [53]. Coupling biomass cultivation with ecosystem restoration and regenerative agriculture practices further enhances the sustainability profile of these systems and can yield additional co-benefits in rural livelihoods and biodiversity [54]. Scaling lignocellulose fermentation systems presents logistical challenges—while decentralized biorefineries benefit from shorter transportation distances for biomass feedstocks, they face financial constraints due to limited economies of scale. On the other hand, large centralized operations contend with high biomass feedstock procurement and transport costs [9]. Recent studies suggest that regional optimization of biorefinery siting—based on feedstock availability, labor access, and proximity to geological storage—can reduce costs and improve environmental outcomes [55]. Future carbon-negative fermentation strategies should prioritize regionally sourced lignocellulosic feedstocks of low value and high quantity, such as agricultural residues, and produce bioproducts of relatively high value. Additionally, establishing fermentation facilities in regions with suitable geological formations for permanent CO₂ sequestration will be necessary to avoid costly CO₂ transport. Emerging low-pressure and passive CO₂ injection technologies may further reduce the infrastructure and energy demands of storage operations, making integration with biorefineries more viable at scale [56]. Finally, novel fermentation pathways for CO₂ removal including sequestration of biogenic carbon in biomaterials should be explored. Examples include microbial conversion of CO₂ into biodegradable plastics, structural biopolymers, and mineralized products, all of which offer potential for long-term carbon storage in durable goods [24,57]. As research in this area advances, fermentation systems may evolve from energy-focused operations into multifunctional carbon removal hubs central to the bioeconomy and climate mitigation landscape.

2.2. Bioplastics: An emerging strategy for BiCRS

To reach net zero by 2050, global emissions need to be reduced at a

rate of 3% per year [58], and by 2040 more than 2500 large-scale CO₂ capture, utilization, and sequestration facilities will likely be required to limit global warming to 2°C [59]. This represents a monumental scaling challenge, as current CCS infrastructure captures only a fraction of that amount globally [60]. Many existing CCS facilities are concentrated in specific industrial hubs, which limits equitable deployment and increases the urgency of exploring complementary carbon removal pathways [61]. Geologic storage projects and CO₂ pipelines can take years to be approved and constructed, and not all biomass exists in close proximity to regions with geologic storage potential. Furthermore, community resistance, permitting delays, and geotechnical limitations often pose barriers to the development of pipeline infrastructure and injection wells, particularly in densely populated or ecologically sensitive regions [10,62]. Given the very large need for CDR, it is important to consider alternatives to traditional gaseous CO₂ capture and removal. These alternatives must prioritize permanence, scalability, environmental safety, and integration with existing industrial systems to meet global climate targets in a socially and economically responsible manner [63]. Bioplastics are highlighted as a novel alternative for CDR [24]. These materials serve as durable carbon sinks when derived from biogenic carbon sources, locking atmospheric CO₂ into long-lived consumer and industrial goods [64]. Bioplastics offer a way to store carbon in a form that is economically valuable and already supported by global infrastructure for manufacturing, packaging, and distribution. Bioplastics, including bio-polyethylene (PE), can be obtained from sustainable sources of biomass and remove carbon dioxide from the atmosphere over their lifetime. One key advantage of non-biodegradable bioplastics including PE is that approximately 75 % of the initial biocarbon remains stored in the product after 100 years [24]; the permanence of PE can increase with proper storage. This long-term carbon retention has led to growing interest in the use of non-biodegradable bioplastics as a verifiable and accountable form of CO₂ sequestration within international carbon markets [64]. The current global production of plastics is estimated between 400 and 500 million metric tonnes per year [65–68] and by 2050 is expected to exceed 1 billion metric tonnes, which offers the bioeconomy an impactful opportunity for atmospheric carbon removal given that the carbon contents of most plastics are higher than 70 % [24, 69]; 1 billion metric tonnes of bioplastic would remove 1.2 billion metric tonnes of CO₂ with a CO₂ removal efficiency of 50%. To contextualize, this scale of removal could offset the annual emissions of a major industrialized nation, such as Japan or Germany, indicating the potential significance of plastics as a carbon sink if global demand is met with biogenic feedstocks and sequestration is ensured [63]. Moreover, because bioplastic production does not necessarily require CO₂ capture, compression, and transport infrastructure, the overall energy use and capital intensity may be lower than traditional BECCS systems. Thus, bioplastics production is an attractive alternative to traditional BECCS due to its potential for carbon sequestration in the final product and avoidance of costly logistical operations including CO₂ gas handling, transporting, and geological injection. In addition, the versatility of plastic products—from packaging to automotive parts to construction materials—allows for distributed carbon storage across a wide array of sectors and timeframes. It's important to highlight that landfilling of durable plastics has a multitude of adverse impacts on the environment and public health, including microplastic generation and contamination of soils and waters, which must be addressed if durable bioplastics are to become a means of atmospheric carbon removal [70–72]. To ensure net climate benefits, carbon sequestered in durable bioplastics must be paired with responsible end-of-life management strategies, including monitored landfilling, engineered burial, or conversion to inert solid forms such as mineralized char. There is also a need to establish global monitoring, reporting, and verification (MRV) protocols that can track carbon permanence across supply chains and enforce standards for storage stability [47]. While bioplastics offer the potential to be a carbon sink, they must be deployed with careful attention to unintended consequences, including land use change, water use, biodiversity loss, and

chemical leaching.

3. Methodology

Eight carbon-negative bioprocesses were designed, modeled, and assessed for techno-economic viability taking into account cradle-to-grave emissions, as explained in Fig. 2. Four primary bioproducts were produced: renewable diesel, ethanol, jet fuel, and PE. For each bioprocess and corresponding primary bioproduct, we assessed two scenarios: with and without co-production of adipic acid (ADA), which is a value-added co-product.

3.1. Biomass feedstocks

The biomass feedstocks selected for this study were based on guidance from the Roads to Removal (RtR) report, which created criteria for pairing biomass conversion technologies with suitable biomass feedstocks [9]. Specifically, fermentation technologies are eligible for biomass feedstocks that have <60 % moisture and minimal inhibitors (e. g. terpenes), according to RtR. Fourteen agricultural and forestry residues were selected as biomass feedstocks for our assessment, with carbon content being the primary tuning parameter in the techno-economic modeling. The specific feedstocks used are as follows, with more information available in the SI: barley straw, citrus residues, corn stover; cotton gin trash, cotton residues, hardwood, lowland logging residues, hardwood, lowland whole trees, hardwood, upland logging residues, hardwood, upland whole trees, non-citrus residues, oats straw, sorghum stubble, sugarcane bagasse, and wheat straw [9]. All of the feedstocks assessed are lignocellulosic in composition and do not compete with current food production [9]. The residues are currently not collected and valorized, and the trees are of small diameter and sustainably grown without land use change. By utilizing these feedstocks, their decomposition and release of GHG emissions are avoided [77]. Feedstocks other than the 14 included in this study could be processed via the fermentation technologies used herein, but costly modifications would likely be necessary.

3.1.1. Pretreatment and enzymatic hydrolysis

One of the most critical factors prior to biomass conversion via fermentation is the breakdown of the biomass into its main components. Several biomass pretreatment technologies have been developed in recent years for this purpose including autohydrolysis, acid hydrolysis, alkaline hydrolysis, and steam explosion. In the present study, deacetylation and mechanical refining (DMR) was selected, due to its cost-effective removal of lignin, polysaccharide retention, and minimal generation of inhibitory compounds [78]. The DMR process enables a more cost-effective hydrolysis compared with other pretreatment technologies including dilute acid hydrolysis [73]. The DMR process has similarities to the soda and kraft pulping processes that use NaOH at elevated temperature and pressure to solubilize lignin while preserving the solid polysaccharides (cellulose and hemicellulose) [73,79]. Thus, the DMR process is highly scalable since it shares many of the same unit operations as the industrial pulping of biomass. After pretreatment, the liquid fraction (mainly lignin) is sent to lignin valorization or combustion depending on the scenario. The residual solid is sent to an enzymatic hydrolysis step to convert the polysaccharides into monomeric sugars. For enzyme hydrolysis, conversion pathways could be developed in a batch or continuous mode operation [80]. In recent years, continuous enzymatic hydrolysis has gained attention [73,80] because this strategy achieves ~96 % of the hydrolysis of polysaccharides to monomeric sugars [73]. A summary of pertinent technical details for the pretreatment and enzyme hydrolysis process is shown in Table 1.

3.2. Lignin upgrading

After pretreatment, the lignin fraction is separated from the solid

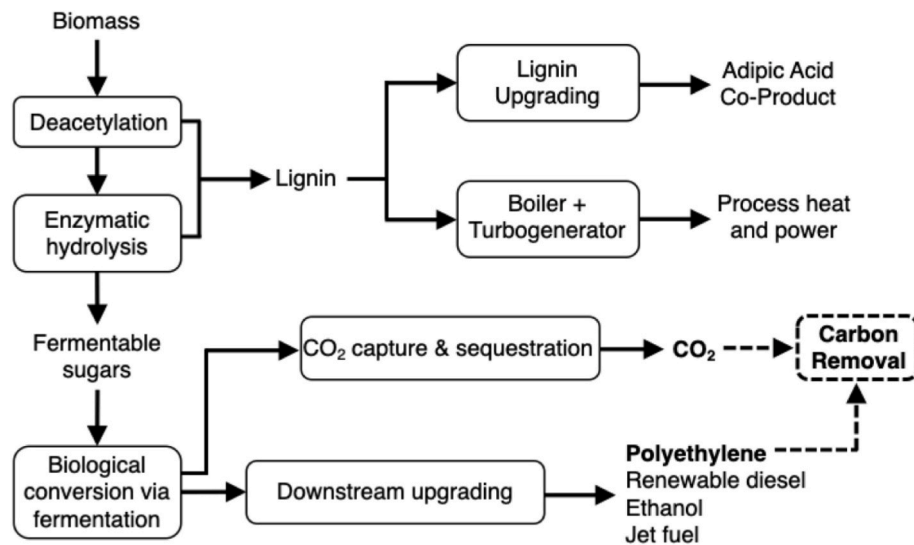


Fig. 2. Simplified flow diagrams of eight bioprocesses assessed for biomass conversion and CO₂ removal: 1) Renewable diesel [73,74]; 2) Ethanol [73–75]; 3) Jet fuel [73–75]; 4) PE [73–76]; 5) Renewable diesel + adipic acid (ADA) [73]; 6) Ethanol + ADA [73–75]; 7) Jet fuel + ADA [73–75]; 8) PE + ADA [73–76].

Table 1

Pertinent technical details for the various processes assessed for biomass conversion and CO₂ removal.

Step	Conditions	Ref
Pretreatment and enzymatic hydrolysis (common for all pathways)		
Deacetylation	92 °C; 90 min; 30 wt% solids loading; 70 mg of NaOH per g of dry biomass 200 kWh/dry metric tonne of biomass	[73, 89]
Enzymatic hydrolysis	50 °C; 7.6 wt% solids loading; 10 mg of protein per g of cellulose >90 % yield of monomeric sugars	[73]
Renewable Diesel		
Fermentation	37 °C; Engineered <i>Clostridium tyrobutyricum</i> ; 0.433 g of butyric acid/g sugar consumed; 2g/L-h productivity	[73]
Upgrading and recovery	365 °C, 180 °C, and 270 °C using ZrO, Niobic acid, and Pt/Al ₂ O ₃ , respectively	[73]
Ethanol		
Fermentation	32 °C; <i>Z. mobilis</i> ; 36 h and 20 wt% solids loading Glucose to ethanol 96 % yield Xylose to ethanol 91 % yield	[74, 75]
Recovery	Two distillation columns achieve 95 % ethanol concentration	[90]
Dehydration	Molecular sieves achieve 99.5 % ethanol concentration	
Jet fuel		
Dehydration	HZSM-5 at 350 °C (0.6 g of ethylene produced/g of ethanol)	[75]
Oligomerization	Ziegler catalyst at 90–120 °C (100 % conversion)	[75]
Hydrogenation	Cobalt molybdenum at 350 °C (100 % conversion)	[75]
Fractionation	Product fractionation by distillation column	
Polymerization		
Polymerization	The conventional process at 100–150 °C at 20 atm	[76]
Drying and Extrusion	The solvent is recovered and solid-phase PE is obtained	[76]

fraction. Lignin could be valorized in several value-added products including vanillin, lignosulfonates, and others [81]. Recently, the production of ADA from lignin has emerged as an attractive option [82,83]. In the present study, we used a conversion process developed by the National Renewable Energy Laboratory (NREL) wherein streams from pretreatment and enzymatic hydrolysis are routed to a reactor that employs caustic soaking and operates at elevated temperatures to further deconstruct the lignin and solubilize oligomers [73]. The soluble liquid product is then routed to bioconversion, which converts soluble lignin and other residual organics to muconic acid through an aerobic

pH-controlled bioconversion. Muconic acid is isolated in high-purity crystals following acid addition and low-pH crystallization. The crystals are redissolved in a carrier solvent and then hydrogenated under mild conditions to ADA, which is subsequently recovered through another crystallizer [73]. A summary of pertinent technical details for the lignin upgrading process is shown in Table 1.

3.3. Sugar conversion

After pretreatment and enzymatic hydrolysis, sugars are converted to value-added products. In the present study, four products were assessed: 1) renewable diesel (RD), 2) ethanol, 3) jet fuel, and 4) PE.

3.3.1. Renewable diesel production

The conversion of sugar to renewable diesel is based on a process wherein the hydrolysate is sent to a fermentation operation to produce butyric acid via an engineered strain of *Clostridium tyrobutyricum*, which is then separated by an extractive membrane system and subsequent distillation. The butyric acid is then sent to a ketonization step, where the acid is upgraded by catalytic conversion and condensation. Finally, the product is recovered by distillation and hydrotreated to remove the remaining oxygen, leaving the isomerized hydrocarbon fuel [73]. Hydrogen gas (H₂) is produced during diesel upgrading at a rate of 44 g per kg of diesel obtained and is completely consumed in lignin upgrading [73]. A summary of pertinent technical details for the sugar to renewable diesel process is shown in Table 1.

3.3.2. Ethanol production

Ethanol fermentation using lignocellulosic feedstocks has typically used the recombinant bacterium *Zymomonas mobilis* or the yeast *Saccharomyces cerevisiae* [84]. In the present study, anaerobic fermentation with *Z. mobilis* was selected after enzymatic hydrolysis [74,75]. The glucose to ethanol yield is more than 96 % of the theoretical maximum, and the xylose to ethanol yield has been demonstrated to be 91 % of the theoretical maximum [75]. After fermentation, ethanol is recovered by distillation and dehydrated via molecular sieves to reach a product purity of more than 99.5 % [75]. A summary of pertinent technical details for the sugar to ethanol process is shown in Table 1.

3.3.3. Jet fuel production via ethanol-to-jet process

In the ethanol-to-jet fuel pathway, the ethanol obtained from distillation is catalytically upgraded into jet fuel via dehydration,

oligomerization, and hydrotreating. [75]. The catalyst used in the dehydration of ethanol can tolerate a low degree of moisture, so the molecular sieve adsorption is not included in the jet fuel pathway. Ethanol dehydration to ethylene can be achieved using several catalysts [85], with HZM-5 being a leading candidate and the catalyst selected for our assessment [86]. Table 1 summarizes the selected conditions for dehydration, wherein ~99 % of the ethanol is converted to ethylene via dehydration. After dehydration, oligomerization and hydrotreating convert ethylene to renewable jet fuel. Oligomerization converts ethylene to linear α -olefins catalytically and hydrogenation is carried out to saturate the remaining double bonds of the olefins [87]. Finally, the products are recovered by distillation obtaining diesel, gasoline, and jet fuel. A summary of pertinent technical details for the ethanol to jet fuel process is shown in Table 1.

3.3.4. Polyethylene (PE) production

The main precursor for bio-PE is bio-ethylene. After ethanol dehydration, bio-ethylene is polymerized into bio-PE using traditional PE polymerization processes [88]. In the present study, we assumed that all ethylene is converted to PE [88]. A summary of pertinent technical details for the ethanol to PE process is shown in Table 1.

3.4. Mass and energy balances

We used published process models with detailed mass and energy flows to construct the integrated processes as shown in Fig. 2. The assumptions across process models were harmonized to ensure accurate comparisons. The initial process operations (biomass handling, pretreatment, and enzymatic hydrolysis) were assessed at the same conditions for all the pathways evaluated. For renewable diesel production, 5.7 kWh of electricity and 0.95 kWh of natural gas are consumed per kg of renewable diesel produced and approximately 32.5 % of total electricity is generated onsite (21.2 MW for corn stover feedstock) [73]. For ethanol production, 1.45 kWh of electricity is consumed per kg of ethanol produced accounting for feedstock handling, pretreatment, and enzymatic hydrolysis operations [73]. In fermentation and ethanol recovery, energy consumption is 4 MJ/kg of ethanol recovered [91]. In the jet fuel pathway, the initial process operations are the same as ethanol production (biomass handling pretreatment, enzymatic hydrolysis, fermentation, and ethanol recovery). For the jet fuel pathway, 6 kg of H₂ per dry metric tonne of biomass is needed in the fuel hydrogenation step [75]. Natural gas represents 78.4 % and electricity represents 21.6 % of total energy consumption in the upgrading of ethanol to ethylene [92]. The total heat demand for upgrading ethanol to jet fuel is 484 kcal/kg ethanol and the total electricity demand is 0.33 kWh/kg ethanol [75]. For ethanol conversion to PE, 9.85 MJ/kg PE is consumed with 78 % of the energy corresponding to natural gas and 22 % corresponding to electricity [88]. An H₂ consumption of 6.1 kg per dry metric tonne of biomass is used for pathways with ADA production [73].

3.5. Life cycle assessment

We conducted a cradle-to-grave life cycle assessment for each pathway to quantify the net CO₂ removed from the atmosphere. Carbon sinks included biomass growth, gaseous CO₂ capture and sequestration, and bioplastic material. GHG emissions were considered in the cultivation, transport, and conversion of biomass, as well as the transport and sequestration of CO₂. Non-CO₂ emissions including CH₄ and N₂O were accounted for in biomass cultivation. Cradle-to-gate carbon flows were quantified using assumptions from the US Department of Energy *Roads to Removal* study, gate-to-gate carbon flows were quantified using detailed process model data available in the literature, and gate-to-grave carbon flows were quantified using assumptions from the US Department of Energy *Roads to Removal* study [9].

The following assumptions were considered in the LCA.

- All direct cradle-to-grave emissions accounted for except the transport of bioproducts from the gate to consumption.
- Indirect emissions from the generation of electricity, hydrogen, and natural gas were accounted for, as well as the indirect emissions from the removal of biomass residues from cropping systems. No embodied emissions were accounted for in the study.
- Carbon content of biomass feedstock is the tuning parameter to differentiate various feedstocks in our modeling.
- CO₂ streams >90 % purity undergo dehydration and compression.
- CO₂ streams <90 % purity undergo monoethanolamine (MEA) separation, dehydration, and compression.
- Carbon durability of sequestered CO₂ and bioplastic is 100 % and 60 % over 100 years, respectively [9]. ADA was not considered a carbon sink due to insufficient LCA data.

There is considerable uncertainty in the carbon durability of PE, but 60 % was selected as a conservative value [9]. Experimental evidence suggests PE degrades into CO₂ via chemical degradation, photodegradation, and biological degradation in landfills; the degradation pathway differs from the anaerobic digestion of organic waste [93–95]. For the biofuel products, GHG emissions from usage are shown in Table S2. Also, PE recycling factors and other life cycle factors assumed are presented in Tables S3 and S4.

3.6. Techno-economic analysis

The techno-economic analysis (TEA) quantified the levelized costs of CO₂ captured and CO₂ removed in USD per t-CO₂; CO₂ captured refers to the quantity of CO₂ physically captured at the biorefinery whereas CO₂ removed refers to the net CO₂ removed from the atmosphere as calculated in the LCA. Variable operating costs, fixed operating costs, and capital investment including return on investment were included in the TEA. Variable operating costs include feedstock, energy, and reagents used, among others and fixed operating costs include factors such as labor and maintenance [9]. Capital investment costs were calculated based on the total direct costs (TDC), including equipment and installation costs, and indirect costs, including land and permitting (see Table S5) [9]. The TDC was calculated based on updated literature [73, 75, 96]. Labor costs were calculated based on the total capital investment (TCI). Pertinent TEA parameters are shown in Table 2 and more detailed TEA data including the itemized direct capital costs are included in the SI (Table S5).

The regression cost equation developed by Psarras et al., [98] was used to quantify the costs of CO₂ capture, and the costs of CO₂ drying and compression were quantified using cost equations in the US

Table 2
Techno-economic assumptions for base case processes [9,97].

Biomass input (dry t/d)	2000
Biomass cost (\$/dry t)	60
Electricity cost (\$/kWh)	0.085
Biomass transport cost (\$/t-km)	0.17
CO ₂ transport cost (\$/t-km)	0.11
Plant utilization	90 %
Interest rate	10 %
Plant lifetime (years)	20
Capital recovery factor	11.75 %
Tax rate (%)	35
Total Capital Investment (TCI) (% of Total Direct Capital)	1.424
Fixed OPEX (\$/year)	4.5 % of TCI
EtOH selling price (\$/gal)	1.62
Diesel selling price (\$/gal)	2.44
Jet fuel selling price (\$/gal)	2.28
PE selling price (\$/kg)	1.21
ADA selling price (\$/gal)	1.72
Gasoline selling price (\$/gal)	2.30
Sodium sulfate selling price (\$/kg)	0.15

Department of Energy Roads to Removal study [9]. Also, CO₂ storage was included in each pathway [99]. For the economic assessment, the total levelized cost of CO₂ removal (\$/tCO₂ removed) was used as the main economic indicator. The economic equations used are described in the SI (Eqs. (1) and (2)).

3.7. Sensitivity analysis

A sensitivity analysis was conducted for each pathway to determine the most influential parameters on the levelized cost of CO₂ removal. We allowed independent process variables to fluctuate $\pm 20\%$ and $\pm 50\%$, and the change in the levelized cost of CO₂ removal was assessed.

4. Results

4.1. Carbon dioxide removal

We quantified carbon dioxide removal potentials based on the biogenic carbon in the feedstock and GHG emissions across the total value chain of the proposed pathways, as described in the methodology section. Using LCA data, the net CO₂ removed per dry t-biomass was calculated for each pathway (Fig. 3). For the 8 pathways we evaluated, the CO₂ removal potential varies between 0.29 and 1.50 tCO₂ removed per dry t-biomass. The PE pathway has the highest potential for CO₂ removal, at 1.1–1.5 tCO₂ removed per dry t-biomass, which is largely due to the removal of CO₂ via both the capture of biogenic CO₂ in production and within the bioplastic material itself. Notably, the PE pathway has the highest GHG emissions but prevails as the most impactful pathway for CO₂ removal due to the carbon stored in the final bioplastic product and not emitted upon usage. In contrast, the jet fuel pathways have the lowest CO₂ removal potential, at 0.29–0.66 tCO₂ removed per dry t-biomass, which is mainly because of the high energy

and H₂ demand as well as the significant CO₂ emissions associated with jet fuel usage. In all pathways, emissions from the process conversion and product usage steps have the highest values (Fig. 3), whereas the emissions from biomass cultivation, biomass transport, and CO₂ transport have the lowest emissions. The yields of main products per dry t-biomass are presented in Table S6.

Fig. 4 shows the difference between CO₂ captured and removed for the eight pathways we evaluated. The CO₂ captured values do not account for GHG emissions, whereas CO₂ removed values do account for GHG emissions. All pathways assessed have the potential to remove CO₂

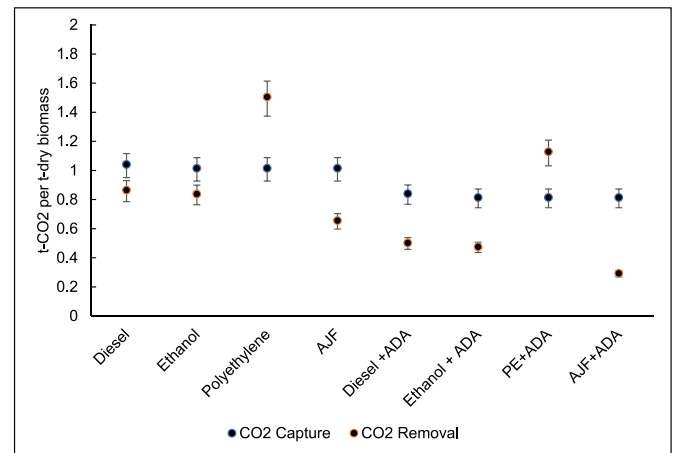


Fig. 4. The quantities of CO₂ captured and removed per metric tonne of dry biomass for eight biomass conversion pathways. The point values represent the corn stover feedstock and the bars represent the distribution including all other feedstocks.

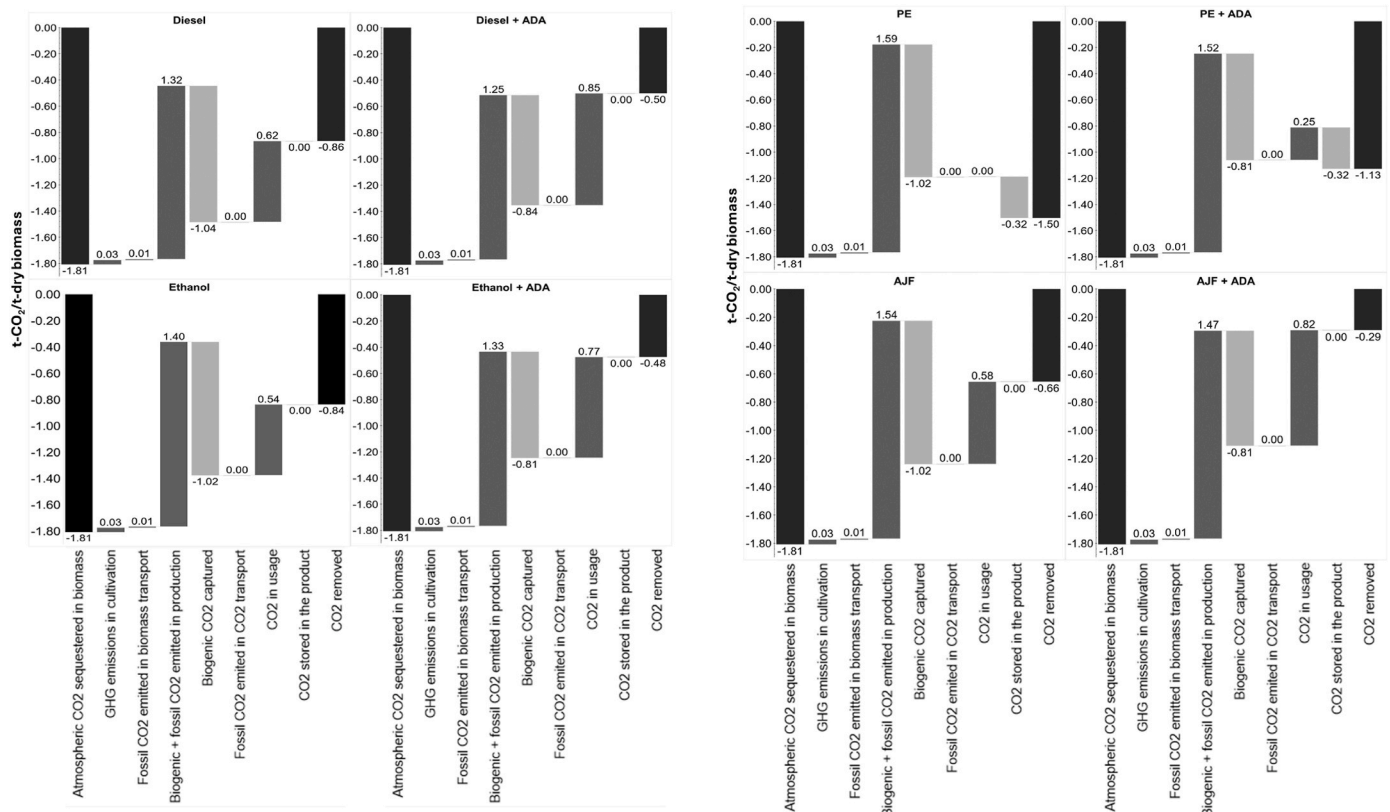


Fig. 3. Waterfall plots showing the CO₂ sinks (black and light grey colors) and CO_{2e} emissions (dark grey) across the life cycle of eight bioproducts. Net CO₂ removal over 100 years is shown at the far right of each plot. All pathways shown in the figure use corn stover as the feedstock.

over their life cycles. Notably, the PE pathways are the only ones with a higher potential for CO₂ removal than CO₂ capture, as shown in (Fig. 4); by dehydrating and polymerizing ethanol into PE, the CO₂ removal potential increases by ~85 % due to the sequestration of biocarbon in the plastic material.

4.2. Energy consumption

The energy consumption for each conversion process is presented in Fig. 5. The PE and jet fuel pathways have the highest total energy consumption, between 1125 and 1480 kWh per dry t-biomass (Fig. 5a); these pathways also have the highest fossil energy consumption. Electricity constitutes the primary form of energy consumption across all pathways (between 55 % and 85 % of total energy) (Fig. 5b). Hydrogen gas is needed in lignin upgrading operations in the ethanol, PE, and jet fuel pathways, and the fossil emissions associated with the production of hydrogen gas are shown in Fig. 5c. In the diesel pathway, H₂ is a byproduct obtained in diesel upgrading, and therefore hydrogen is produced onsite for this particular pathway. As shown in Fig. 5c, the PE pathways have relatively high emissions of fossil CO₂ from grid electricity, natural gas heating, and H₂ production. On the other hand, the PE pathways have relatively high extents of carbon removal, as shown in Fig. 4, which is due to the significant quantities of biogenic carbon sequestered in the bioplastic material. Although the biofuel pathways emit less fossil CO₂ in their production, relative to PE, the quantity of biogenic carbon emitted as CO₂ upon fuel combustion greatly exceeds the differential quantity of fossil CO₂ emissions.

4.3. Economic assessment

Of the eight pathways we evaluated, the highest capital investment costs were associated with the jet fuel pathway due largely to the capital-

intensive fuel upgrading units. The lowest capital investment cost corresponds to the ethanol pathway, due largely to the simplistic nature of the process (Table S5). The PE + ADA pathway represents the lowest levelized cost of CO₂ removal due to the high CO₂ removal efficiency (Fig. 6). The jet fuel pathway has the highest levelized cost of CO₂ removal due to high energy demand and a low CO₂ removal efficiency (Fig. 6). The levelized costs of gaseous CO₂ capture vary between ~\$85 for PE and ~\$400 for jet fuel (Fig. 6). The costs of CO₂ removal are higher than the costs of CO₂ capture for all biofuel pathways, as expected. The cost of CO₂ removal for the PE pathway is slightly lower

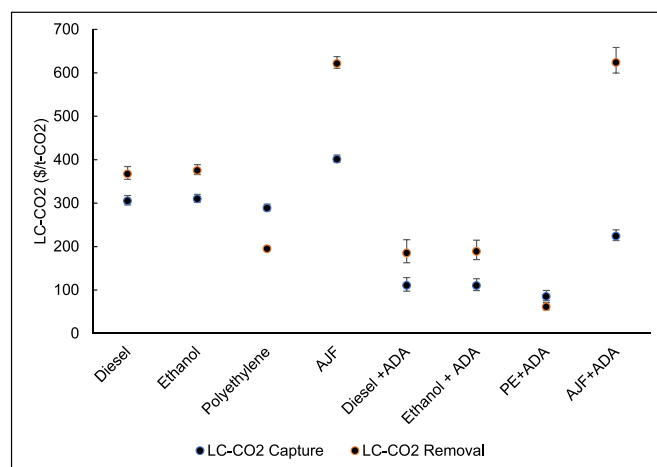


Fig. 6. Levelized costs of CO₂ (LC-CO₂) capture and removal for the eight biomass conversion pathways. The point values represent the corn stover feedstock and the bars represent the distribution including all other feedstocks.

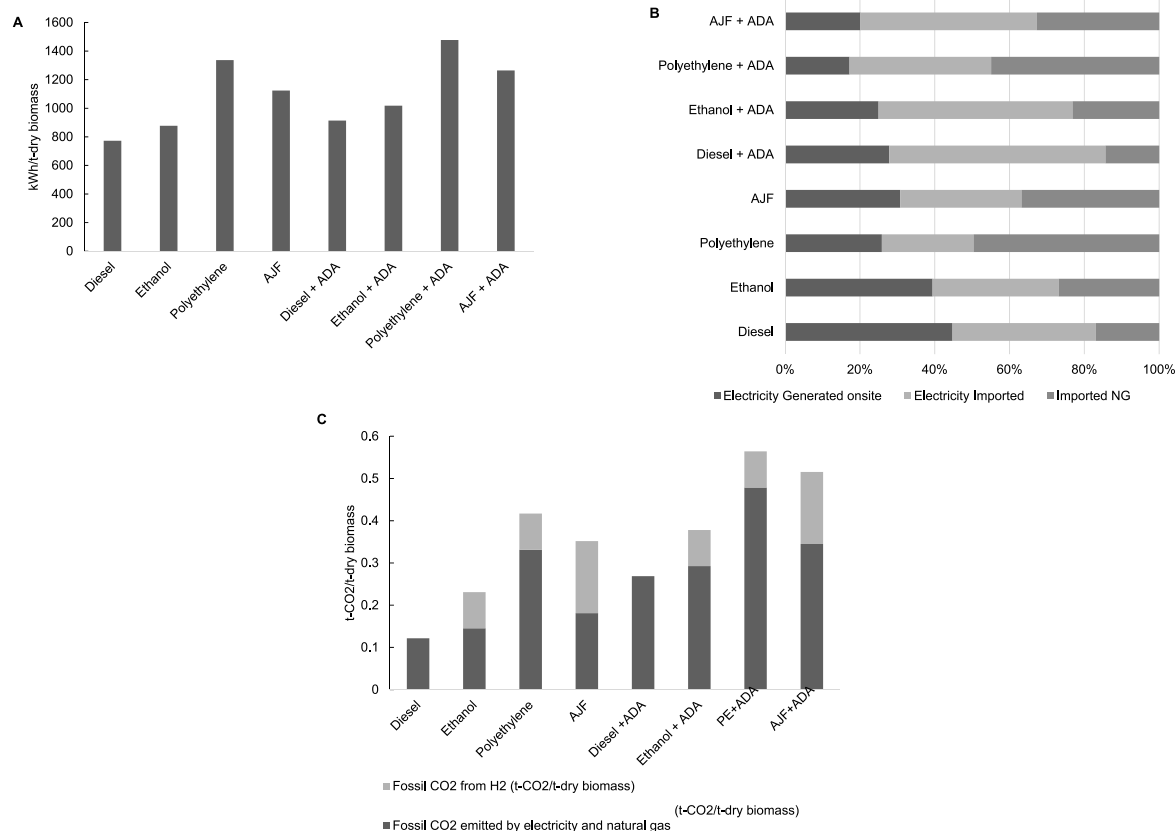


Fig. 5. A) Total energy consumption per dry t-biomass for each pathway, B) Percentages of energy type consumed for each pathway, and C) Quantities and sources of fossil CO₂ emissions for each pathway. All pathways shown in the figure use corn stover as feedstock.

than the cost of CO₂ capture due to the inclusion of CO₂ stored in the bioplastic product. Based on the reported literature, common costs of CO₂ removal via BECCS pathways are \$100 – \$500 per metric tonne of CO₂ [100], which is in the range of the values obtained in our study. The exception is for jet fuel pathways, which have higher costs due to the

high capital intensity and low CO₂ removal efficiency. A significant challenge to the large-scale deployment of BECCS is its competition with low-cost renewable electricity generation—an issue that does not affect bioplastics or biofuels (for transportation) [24]. Similar to BECCS, direct air capture (DAC) is a widely utilized method of engineered carbon

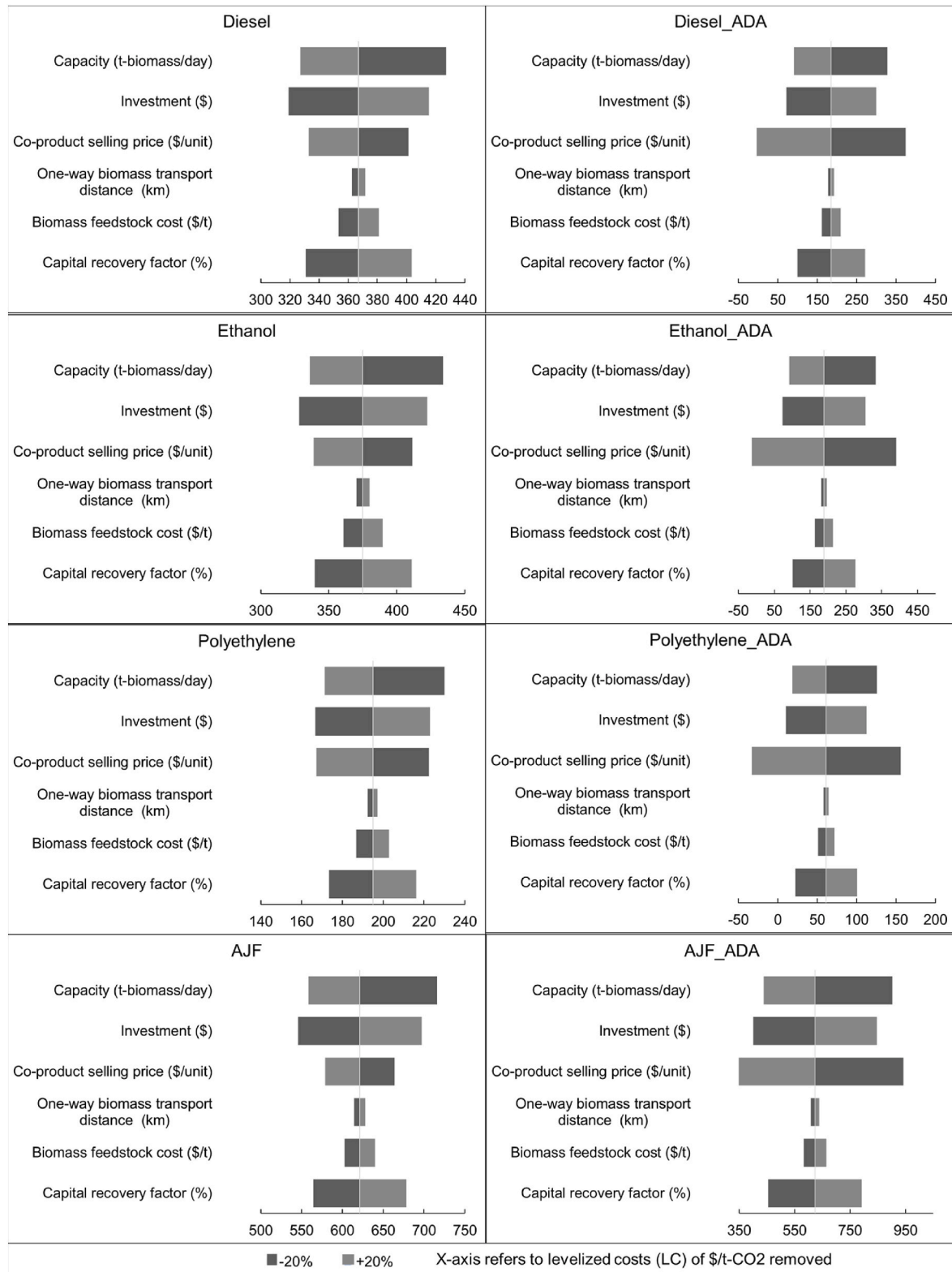


Fig. 7. Tornado plots showing the sensitivity of various techno-economic parameters on the levelized cost of CO₂ removal for eight biomass conversion pathways. All pathways shown in figure use corn stover as feedstock.

removal, offering the advantages of remote siting and centralized operation while facing the drawback of high energy demand [9]. Levelized costs of DAC for nth of a kind facilities are \$250 to \$600 per metric tonne of CO₂ [101], which are considerably higher than some of the fermentation pathways presented herein.

4.4. Sensitivity analysis

In the sensitivity analysis, we varied several techno-economic parameters to understand their influence on the levelized cost of CO₂ removal (Fig. 7). In pathways without ADA as a co-product, biomass capacity, capital investment, co-product selling price, and capital recovery factor are the most influential parameters on the levelized cost of CO₂ removal. The fermentation processes were originally designed for relatively large scales of operation (>1500 dry metric tonnes of biomass per day) and thus are not well suited for small, modular scales of operation, evident by the high costs of removal for the lower capacities in Fig. 7 [73]. Fermentation technologies are generally highly capital-intensive, and thus economies of scale are important. The feedstocks selected for this study are centrally located in high abundance and the costs associated with collection and transport are relatively low due to the low moisture and uniform composition, thereby making them suitable for large scale fermentation processes [9].

In pathways with ADA as a co-product, the co-product selling price is the most influential variable, with biomass capacity and investment costs being less important. For all evaluated pathways, biomass feedstock costs and biomass transport costs have relatively little influence. The levelized costs of CO₂ removal for the ethanol and diesel with ADA pathways have the potential to be near-zero when the co-product selling price is increased by 20%. Notably, the levelized cost of CO₂ removal for the PE with ADA pathway is negative at a high co-product selling price, indicating that additional profit is generated and there is potential to increase the return on investment in the capital recovery factor.

Overall, the PE with ADA is the most compelling pathway we evaluated due to the ability to reach near-zero CO₂ removal costs with fluctuation in capacity, investment cost, and capital recovery factor. The jet fuel pathways show some economic improvement when biomass capacity is increased and capital costs are decreased, but overall the costs remain high (>\$350 per tCO₂ removed) due to the low CO₂ removal efficiency. Fig. S2 shows an additional sensitivity analysis using extreme values ($\pm 50\%$), from which similar trends are observed. Notably, increasing the capacity by 50% to 3000 dry metric tonnes biomass per day results in a levelized cost of CO₂ removal near zero for the PE + ADA pathway, meaning that the production and sale of PE and ADA would enable economic viability without the need for revenue from the sale of carbon removal credits. In such a scenario, the business model is much more risk averse since fluctuations in carbon removal markets or sudden changes in policy incentives would not impact economic viability. In general, business models that provide high quality carbon removal, but aren't reliant on the revenue generated from the sale carbon removal credits, warrant further investigation. Globally, there are a multitude of policies that influence the production of biofuels and bioplastics, but, to the best of our knowledge, the majority of such policies do not incentivize BiCRS [70,102]. Policymakers should stay informed of the growing evidence that shows BiCRS to be a scalable and low-cost approach to mitigating climate change while also creating domestic jobs in the bioeconomy. Our sensitivity analyses confirm the capital-intensive nature of industrial fermentation and recommend increasing economies of scale and/or finding ways to reduce capital costs as the primary means of reducing the costs of CO₂ removal. Further, high-value products that store biogenic carbon, such as PE, can greatly reduce the cost of CO₂ removal.

5. Discussion

We assessed various biochemical pathways for the production of

carbon-negative biofuels and bioplastics to quantify and compare the potential impacts and costs of CO₂ removal. Removing CO₂ from the atmosphere via biofuel pathways was found to be more expensive and less impactful than the bioplastic pathways due to the return of biogenic carbon to the atmosphere as CO₂ upon biofuel combustion. Bioplastics stabilize biogenic carbon thereby decreasing cost and increasing the impact of CO₂ removal. Biofuels are major global commodities with production expected to increase ~30% by 2030 to avoid fossil carbon emissions and stay in line with net-zero targets [103]. The use of biofuels has traditionally been a strategy of CO₂ avoidance and not CO₂ removal, but, as shown by our analysis, CO₂ avoidance plus CO₂ removal is possible via net negative biofuels. However, the costs of CO₂ removal via biofuel pathways are high (\$185 - \$625 per tCO₂) due to the little amount of CO₂ sequestered relative to the total amount of CO₂ emitted over the biofuel life cycle (see Fig. 6). In our analysis, we did not account for the avoided fossil CO₂ emissions from the use of biofuels in our CO₂ removal cost calculations, because avoided emissions do not contribute to CO₂ removal. Of course, avoiding future emissions of fossil CO₂ through the use of biofuels is important when one assumes a counterfactual trajectory involving continued use of fossil fuels. Atmospheric CO₂ removal is a physical process that is assessed without counterfactuals, and thus avoided emissions were not included in this study.

The standalone need for atmospheric CO₂ removal is important as it serves as a mechanism to account for legacy fossil emissions already in the atmosphere and hard-to-abate emissions, including methane and nitrous oxide [9]. Thus, CO₂ removal is needed regardless of the need for avoiding future fossil CO₂ emissions. This study determined that among common, industrial biochemical conversion technologies, fermentation and subsequent upgrading of lignocellulosic feedstocks to PE bioplastics is an attractive method for CO₂ removal with costs ranging from \$61 to \$195 per tCO₂ removed (with and without ADA, respectively); the lower end of the CO₂ removal cost range via PE involves the sale of ADA as a high value co-product. The costs of CO₂ removal via PE with and without ADA are considerably lower than the costs of CO₂ removal via biofuels with and without ADA (\$320 - >\$700 per tCO₂), as shown in Fig. 7. Adipic acid was selected as the value-added co-product from lignin because of its high value. Notably, adipic acid can be upgraded further into nylon 6,6, which has a larger market demand than adipic acid [104]. Aside from adipic acid, there is a multitude of different chemical co-products that could be produced from lignin, such as phenol derivatives including vanillin and BTX compounds (benzene, toluene, and xylene). Further, various biomaterials can be made from lignin, including foams, resins, and activated carbon [81,105].

PE is a major commodity that is sold into highly competitive markets, and thus carbon-negative bio-PE will only scale if its selling price is competitive with incumbent fossil carbon-derived PE [106]. Herein, we determine the minimum selling price of carbon removal (\$/tCO₂), which is equal to levelized cost, that enables economic viability for a business that is selling bio-PE at a competitive market price. In other words, the proposed bio-PE product will be economically viable in existing markets if buyers of carbon removal credits are secured. The market size for ADA is small compared to PE, and thus the majority of the potential climate benefit exists in the production of PE without ADA at a cost of \$195 per tCO₂ removed. Therefore, fermentation to PE is a less attractive pathway for carbon removal when compared to thermochemical conversion pathways, including gasification, pyrolysis, and combustion [9]. Gasification of sustainably sourced biomass to hydrogen and sequestered CO₂ is particularly more competitive than fermentation to PE. The cost of CO₂ removal via gasification has been shown to be ~\$75 per metric tonne CO₂ removed [9]. However, carbon-negative biomass gasification requires a substantial increase in consumer demand for hydrogen and costly infrastructure to transport the hydrogen and CO₂ products. Carbon-negative fermentation to PE leverages existing market demand for plastics and requires less infrastructure since sequestration of PE is less complex than CO₂.

The global annual demand for PE exceeds 100 million metric tonnes,

which would require 600 million dry metric tonnes of biomass and sequester 252 million metric tonnes of CO₂ per year if met by the bio-PE technology presented herein [107]. In the US alone, there is capacity to sustainably harvest over 1 billion dry metric tonnes of lignocellulosic biomass per year, and thus there is sufficient biomass availability to fully meet global PE demand [9]. The concept of intentionally creating non-degradable bioplastics for CO₂ removal may not be readily accepted by society given the multi-decade emphasis towards circularity and recycling of plastic materials, and further studies should be conducted to understand trade-offs and to educate the public of such trade-offs [108]. Life cycle assessments comparing bio-based and fossil-based PE indicate that other impact factors not related to fossil energy consumption and net GHG emissions are moderately more favorable for fossil-based PE, as shown in Table 3. All of the biomass feedstocks used in the studies referenced in Table 3 were food crops, such as sugarcane, and thus the life cycle impacts should be improved with the use of non-food, lignocellulosic feedstocks. Nonetheless, bio-based PE may pose greater risks in terms of acidification, eutrophication, photochemical ozone formation, land use, and water consumption. Ongoing research is essential to assess the trade-offs between the climate benefits and potential environmental impacts of carbon-negative bioplastic production [68, 70–72].

6. Conclusions

Eight different fermentation-based biomass conversion pathways were assessed for their techno-economic potential to remove CO₂ from the atmosphere. The bioproducts assessed included biofuels and biomaterials, with PE being the lowest cost option for CO₂ removal due to the capture of biogenic CO₂ from fermentation and sequestration of biocarbon in the plastic material itself. Among the biofuel pathways, ethanol and diesel pathways presented lower levelized costs of CO₂ removal compared with the jet fuel pathways, mainly due to high capital costs and life cycle emissions associated with the jet fuel pathways. The inclusion of ADA as a byproduct significantly decreased the cost of CO₂ removal for each pathway due to the additional revenue. ADA was not considered as a carbon sink due to insufficient LCA data, but future work should assess its potential for CO₂ removal. Sensitivity assessment indicated that the inclusion of coproducts can considerably improve the levelized cost of CO₂ removal, and that biomass capacity, investment, and capital recovery factors are important factors. When optimizing for CO₂ removal, bioproducts that can sequester carbon in the final product, such as stable bioplastics, are promising options. Future work should investigate the potential advantages and disadvantages of scaling up bioplastic production for large-scale CO₂ removal, and the choice to include or exclude avoided fossil CO₂ emissions should be examined further.

Author contributions

W.J.S. and N.C. conceptualized the study; N.C. conducted the analysis; W.J.S., N.C., C.D.S., and J.P. developed the methodology and interpreted results; N.C. wrote the original draft, W.J.S. reviewed and edited; W.J.S. and N.C. wrote the paper with input from all co-authors.

Proofreading

During the preparation of this work the authors used Chat GPT in order to proofread writing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Table 3

Comparison of life cycle impacts for bio-based and fossil-based polyethylene.

Life cycle impact	Units	Bio-PE	Fossil-PE	Ref.
Acidification	g SO ₂ -eq/kg LDPE	13.9	9.9	[109]
Eutrophication	g PO ₄ -eq/kg LDPE	2.1	1.1	[109]
Photochemical ozone creation potential	g ethylene/kg LDPE	2.9	2.7	[109]
Land use	m ² /kg PE	4.3–4.8	–	[110]
Water use	m ³ water/kg PE	1.39 – 1.69	0.45	[111, 112]

William Joe Sagues reports financial support was provided by US Department of Energy. Corinne D. Scown reports a relationship with Cyklos Materials that includes: equity or stocks. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.rser.2025.115714>.

Data availability

Data will be made available on request.

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